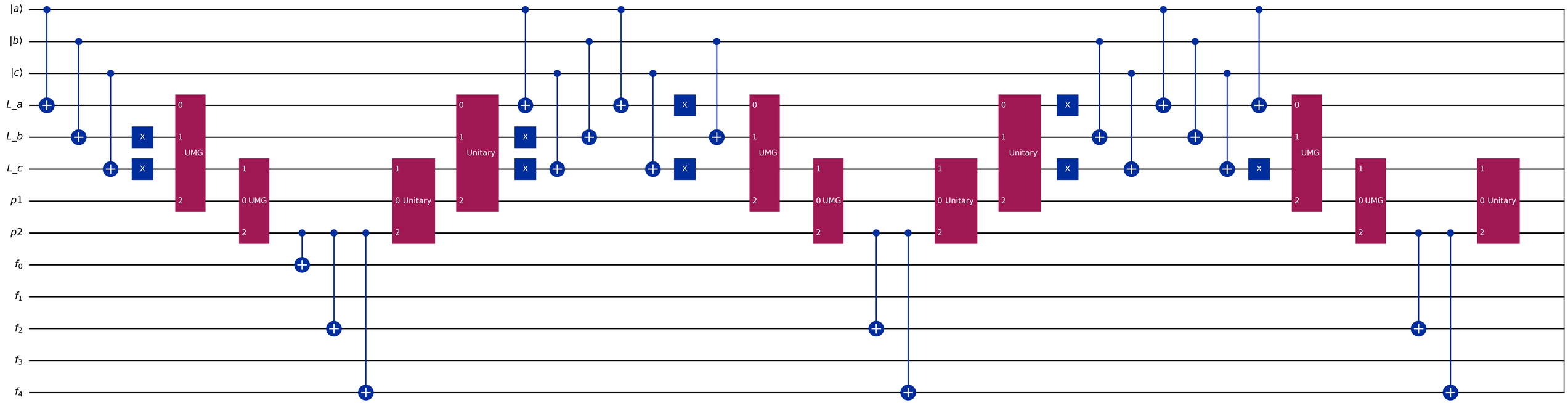
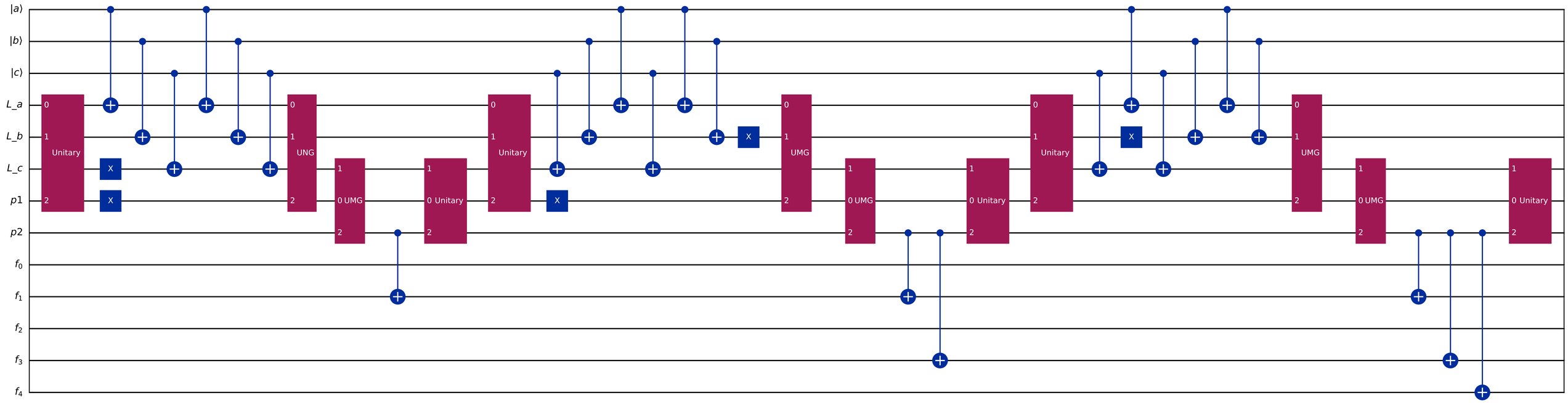

|a⟩
|b⟩
|c⟩
L_a
L_b
L_c
p1
p2
f0
f1
f2
f3
f4
X
0
1
UMG
2
1
0 UMG
2
1
0 Unitary
2
0
1
Unitary
2
X
X
X
X
0
1
UMG
2
1
0 UMG
2
1
0 Unitary
2
0
1
Unitary
2
X
X
X
X
0
1
UMG
2
1
0 UMG
2
1
0 Unitary
2


|a⟩
|b⟩
|c⟩
L_a
L_b
L_c
p1
p2
f0
f1
f2
f3
f4
0
1
Unitary
2
X
X
0
1
UNG
2
1
0 UMG
2
1
0 Unitary
2
0
1
Unitary
2
X
X
0
1
UMG
2
1
0 UMG
2
1
0 Unitary
2
0
1
Unitary
2
X
X
0
1
UMG
2
1
0 UMG
2
1
0 Unitary
2

|a⟩
|b⟩
|c⟩
L_a
L_b
L_c
p1
p2
f0
f1
f2
f3
f4
0
1
Unitary
2